

LAB 2D: Queue it up!
Response Sheet

Directions: Record your responses to the lab questions in the space provided.

Where we left off

Back to songs

- (1) Write and run code creating a *playlist* of songs containing 30 "rap" songs, 23 "country" songs and 47 "rock" songs.

- (2) Write and run code to randomly select a single song 50 times. Then use your 50 sample draws to estimate the probability of choosing a *rap* song.

- (3) Write a sentence comparing your estimated probability to the "true" probability.

With or Without?

- (4) Write and run code taking a sample of size 100 from our playlist of songs *without replacement*. Assign this sample the name *without*.

- (5) Run `tally(without)` and describe the output.

- (6) Does something similar happen if you sample *with replacement*?

- (7) What happens if `size = 101` and `replace = FALSE`?

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Sample with? Or without?

Imagine the following two scenarios.

1. You have a coin with two sides: *Heads* and *Tails*. You're not sure if the coin is fair and so you want to estimate the probability of getting a *Head*.
 2. A child reaches into a candy jar with 10 *strawberry*, 50 *chocolate* and 25 *watermelon* candies. The child is able to grab three candies with their hand and you're interested in the probability that all three candies will be chocolate.
- (8) Which of these scenarios would you sample *with replacement* and which would you sample *without replacement*? Why?

- (9) Write down the line of code you would run to sample from the candy jar. Name the candy jar data that you create candies.

Simulations at work

- (10) Write and run code using the `do` function to perform 10 simulated samples of size 2, without replacement and *assign* the output of the simulated samples the name `draws` and then `View` your file. Use `set.seed(1)`.
- (11) What are the variable names? What happened in the first simulated sample? Did any of your 10 simulated samples contain two *rap* songs?

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Simulations and probability

Counting similar outcomes

- (12) Let's break down the code above by running each part of the code one piece at a time. As you run each line of code below describe the output.

```
draws == "rap"
```

```
rowSums(draws == "rap")
```

```
mutate(draws, nrap = rowSums(draws == "rap"))
```

Counting other outcomes

Step 1: Creating a subset

- (13) Fill in the blanks below to:
1. Create a subset of our simulations when both draws were "rap" songs.
 2. Count the number of rows in this subset.
 3. And divide by the total number of repeated simulations.

```
draws_sub <- filter(draws, _____ == "rap", _____ == "rap")
```

```
nrow(_____) / _____
```

Estimating probabilities

- (14) Write and run code performing 500 simulated samples of selecting 2 songs from a playlist of 30 "rap", 23 "country" and 47 "rock" songs. You might consider running `set.seed()` so that your results can be reproduced.

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(15) Calculate and write down the estimated probabilities for the following situations:

1. You draw two "rap" songs.

2. You draw a "rap" song in the first draw and a "country" song in the 2nd.

(16) Create a histogram that displays the number of times a "rap" song occurred in each simulation. How often were zero rap songs drawn? A single rap song? Two rap songs?

On your own

(17) If we draw 5 songs from a playlist of 30 rap, 23 country and 47 rock songs, how does the estimated probability of all 5 songs being rap songs change if we draw the songs with or without replacement?

(18) Describe how the distribution of the number of *rap* songs changes depending on if we use replacement or not.